COM105 Engineering Physics (I)

Tutorial 1

1. Starting with the definition 1 in. = 2.54 cm, find the number of (a) kilometers in 1.00 mile and (b) feet in 1.00 km.

2. According to the label on a bottle of salad dressing, the volume of the contents is 0.473 liter (L). Using only the conversions 1 L = 1000 cm^3 and 1 in. = 2.54 cm, express this volume in cubic inches.

3. How many nanoseconds does it take light to travel 1.00 ft in vacuum? (The speed of light is v = 3×10^8 m/s)

4. A certain fuel-efficient hybrid car gets gasoline mileage of 55.0 mpg (miles per gallon). (a) If you are driving this car in Europe and want to compare its mileage with that of other European cars, express this mileage in km/L. Use the conversion factors in Appendix E. (b) If this car's gas tank holds 45 L, how many tanks of gas will you use to drive 1500 km?

5. The rest energy *E* of an object with rest mass *m* is given by Albert Einstein's famous equation $E = mc^2$, where *c* is the speed of light in vacuum. Find E for an electron for which (to three significant figures) m = 9.11×10^{-31} kg. The SI unit for E is the joule (J); 1 J = 1 $kg.m^2/s^2$.

6. A useful and easy-to-remember approximate value for the number of seconds in a year is $\pi \times 10^7$. Determine the percent error in this approximate value. (There are 365.24 days in one year.)

7. Express each approximation of π to six significant figures: (a) 22/7 and (b) 355/113. (c) Are these approximations accurate to that precision?

8. For the vectors **A** and **B** in Fig. 1.24, use the method of components to find the magnitude and direction of (a) the vector sum **A** + **B**; (b) the vector sum **B** + **A**; (c) the vector difference **A** – **B** and (d) the vector difference **B** – **A**.

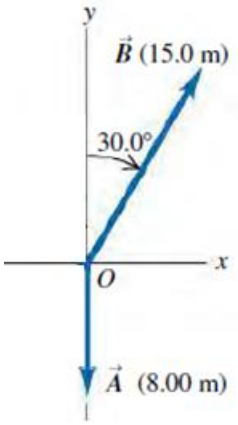


Figure 1.24

9. For the vectors **A**, **B**, and **C** in Fig. 1.25, find the scalar products (a) A • B; (b) B • C; (c) A • C.

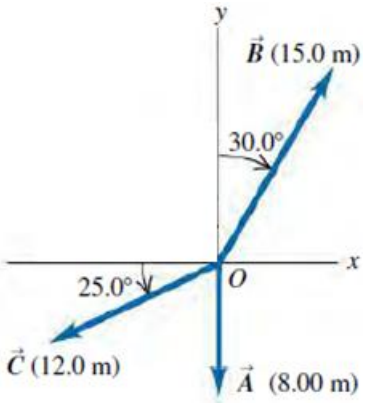


Figure 1.25

10. For the two vectors **A** and **D** in Fig. 1.26, find the magnitude and direction of (a) the vector product **A** × **D** and (b) **D** × **A**.

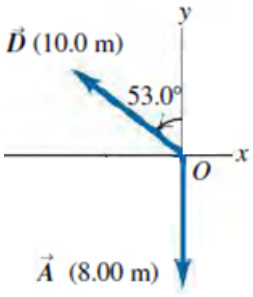


Figure 1.26

11. Given the two displacements

$\vec{D} = (6.00\hat{\imath} + 3.00\hat{\jmath} - 1.00\hat{k})$ m and

$\vec{E} = (4.00\hat{\imath} - 5.00\hat{\jmath} + 8.00\hat{k})$ m

Find the magnitude of the displacement 2**D** – **E**.

12. You are given two vectors **A** = –3.00*i* + 6.00*j* and **B** = 7.00*i* + 2.00*j*. Let counterclockwise angles be positive. (a) What angle does **A** make with the +x-axis? (b) What angle does **B** make with the +x-axis? (c) Vector **C** is the sum of **A** and **B**, so **C** = **A** + **B**. What angle does **C** make with the +x-axis?

13. Given two vectors A = –2.00*i* + 3.00*j* + 4.00*k* and B = 3.00*i* + 1.00*j* – 3.00*k*, (a) find the magnitude of each vector; (b) use unit vectors to write an expression for the vector difference **A** – **B**; and (c) find the magnitude of the vector difference **A** – **B**. Is this the same as the magnitude of **B** – **A**? Explain.

14. Two ropes in a vertical plane exert equal-magnitude forces on a hanging weight but pull with an angle of 72.0° between them. What pull does each rope exert if their resultant pull is 372 N directly upward?

15. A sailor in a small sailboat encounters shifting winds. She sails 2.00 km east, next 3.50 km southeast, and then an additional distance in an unknown direction. Her final position is 5.80 km directly east of the starting point. Find the magnitude and direction of the third leg of the journey. Draw the vector-addition diagram and show that it is in qualitative agreement with your numerical solution.

16. (a) What are the x- and y-components of vector **D** in Fig. 1.19a? The magnitude of the vector is D = 3.00 m, and angle $\alpha = 45°$.
(b) What are the x- and y-components of vector **E** in Fig. 1.19b? The magnitude of the vector is E = 4.50 m, and angle $\beta = 37.0°$.

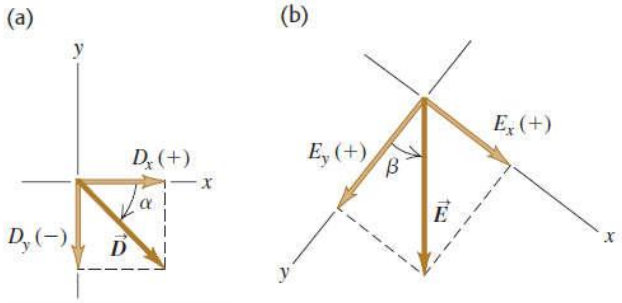


Figure 1.19

17. Three players on a reality TV show are brought to the center of a large, flat field. Each is given a meter stick, a compass, a calculator, a shovel, and (in a different order for each contestant) the following three displacements:

A: 72.4 m, 32.0° east of north
B: 57.3 m, 36.0° south of west
C: 17.8 m due south

The three displacements lead to the point in the field where the keys to a new Porsche are buried. Two players start measuring immediately, but the winner first calculates where to go. What does she calculate?

18. Find the scalar product **A.B** of the two vectors in Fig. 1.28. The magnitudes of the vectors are A = 4.00 and B = 5.00.

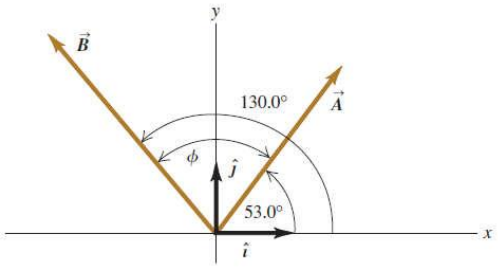


Figure 1.28

APPENDIX E

UNIT CONVERSION FACTORS

Length
1 m = 100 cm = 1000 mm = $10^6\ \mu$m = 10^9 nm
1 km = 1000 m = 0.6214 mi
1 m = 3.281 ft = 39.37 in.
1 cm = 0.3937 in.
1 in. = 2.540 cm
1 ft = 30.48 cm
1 yd = 91.44 cm
1 mi = 5280 ft = 1.609 km
1 Å = 10^{-10} m = 10^{-8} cm = 10^{-1} nm
1 nautical mile = 6080 ft
1 light-year = 9.461×10^{15} m

Area
1 cm^2 = 0.155 $in.^2$
1 m^2 = 10^4 cm^2 = 10.76 ft^2
1 $in.^2$ = 6.452 cm^2
1 ft^2 = 144 $in.^2$ = 0.0929 m^2

Volume
1 liter = 1000 cm^3 = 10^{-3} m^3 = 0.03531 ft^3 = 61.02 $in.^3$
1 ft^3 = 0.02832 m^3 = 28.32 liters = 7.477 gallons
1 gallon = 3.788 liters

Time
1 min = 60 s
1 h = 3600 s
1 d = 86,400 s
1 y = 365.24 d = 3.156×10^7 s

Angle
1 rad = 57.30° = 180°/π
1° = 0.01745 rad = π/180 rad
1 revolution = 360° = 2π rad
1 rev/min (rpm) = 0.1047 rad/s

Speed
1 m/s = 3.281 ft/s
1 ft/s = 0.3048 m/s
1 mi/min = 60 mi/h = 88 ft/s
1 km/h = 0.2778 m/s = 0.6214 mi/h
1 mi/h = 1.466 ft/s = 0.4470 m/s = 1.609 km/h
1 furlong/fortnight = 1.662×10^{-4} m/s

Acceleration
1 m/s^2 = 100 cm/s^2 = 3.281 ft/s^2
1 cm/s^2 = 0.01 m/s^2 = 0.03281 ft/s^2
1 ft/s^2 = 0.3048 m/s^2 = 30.48 cm/s^2
1 mi/h · s = 1.467 ft/s^2

Mass
1 kg = 10^3 g = 0.0685 slug
1 g = 6.85×10^{-5} slug
1 slug = 14.59 kg
1 u = 1.661×10^{-27} kg
1 kg has a weight of 2.205 lb when g = 9.80 m/s^2

Force
1 N = 10^5 dyn = 0.2248 lb
1 lb = 4.448 N = 4.448×10^5 dyn

Pressure
1 Pa = 1 N/m^2 = 1.450×10^{-4} $lb/in.^2$ = 0.0209 lb/ft^2
1 bar = 10^5 Pa
1 $lb/in.^2$ = 6895 Pa
1 lb/ft^2 = 47.88 Pa
1 atm = 1.013×10^5 Pa = 1.013 bar
= 14.7 $lb/in.^2$ = 2117 lb/ft^2
1 mm Hg = 1 torr = 133.3 Pa

Energy
1 J = 10^7 ergs = 0.239 cal
1 cal = 4.186 J (based on 15° calorie)
1 ft · lb = 1.356 J
1 Btu = 1055 J = 252 cal = 778 ft · lb
1 eV = 1.602×10^{-19} J
1 kWh = 3.600×10^6 J

Mass–Energy Equivalence
1 kg ↔ 8.988×10^{16} J
1 u ↔ 931.5 MeV
1 eV ↔ 1.074×10^{-9} u

Power
1 W = 1 J/s
1 hp = 746 W = 550 ft · lb/s
1 Btu/h = 0.293 W

APPENDIX F

NUMERICAL CONSTANTS

Fundamental Physical Constants*

Name	Symbol	Value
Speed of light in vacuum	c	2.99792458×10^{8} m/s
Magnitude of charge of electron	e	$1.602176565(35) \times 10^{-19}$ C
Gravitational constant	G	$6.67384(80) \times 10^{-11}$ N · m^2/kg^2
Planck's constant	h	$6.62606957(29) \times 10^{-34}$ J · s
Boltzmann constant	k	$1.3806488(13) \times 10^{-23}$ J/K
Avogadro's number	N_A	$6.02214129(27) \times 10^{23}$ molecules/mol
Gas constant	R	$8.3144621(75)$ J/mol · K
Mass of electron	m_e	$9.10938291(40) \times 10^{-31}$ kg
Mass of proton	m_p	$1.672621777(74) \times 10^{-27}$ kg
Mass of neutron	m_n	$1.674927351(74) \times 10^{-27}$ kg
Magnetic constant	μ_0	$4\pi \times 10^{-7}$ Wb/A · m
Electric constant	$\epsilon_0 = 1/\mu_0 c^2$	$8.854187817... \times 10^{-12}$ C^2/N · m^2
	$1/4\pi\epsilon_0$	$8.987551787... \times 10^{9}$ N · m^2/C^2